

Non-Terrestrial Networks and Space Security

Dr. Linke Guo
Clemson University

Dr. Alex Sprintson
George Mason University
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AI disclosure – AI was used to polish the presentation.

Working document

<https://tinyurl.com/satcsat>



Rationale

Key drivers and security challenges of modern non-terrestrial architectures



Foundational Role

Non-terrestrial networks (NTNs) have evolved into critical infrastructure for both defense and civilian ecosystems.



Distributed Edge

Emerging space-based networks may eventually host distributed data centers directly within satellite constellations.



Persistent Threats

The inherently wireless nature of these systems exposes them to persistent and increasingly sophisticated security threats.



Cross-Domain

Tight coupling between space assets and terrestrial systems introduces complex, unique cross-domain dependencies.

NTN Networks

Key elements of Non-Terrestrial Network architectures



LEO/MEO/GEO

Multi-orbit satellite constellations providing global coverage and latency-optimized communications.



HAPS / UAVs

High-Altitude Platform Stations and drones serving as airborne base stations for rapid deployment.



Direct-to-Device

Direct satellite-to-phone links enabling unmodified standard devices to access space-based connectivity.



Integrated Systems

Seamlessly integrated terrestrial and non-terrestrial infrastructures forming cohesive hybrid networks.



GNSS / PNT

Critical satellite-based positioning, navigation, and timing services essential for global coordination.

Rationale (Cont.)

New attack surfaces/vectors due to:

Vast Areas

Expansive coverage requirements and remote operations across globally distributed segments.

Weak Signals

Heavy reliance on long-range radio links and highly vulnerable, broad signal footprints, interference from space weather events.

High Mobility

Fast-moving nodes requiring rapid handovers and dynamic network topology adaptations.

Autonomy

Satellites must authenticate users and make critical authorization decisions without contacting home networks.

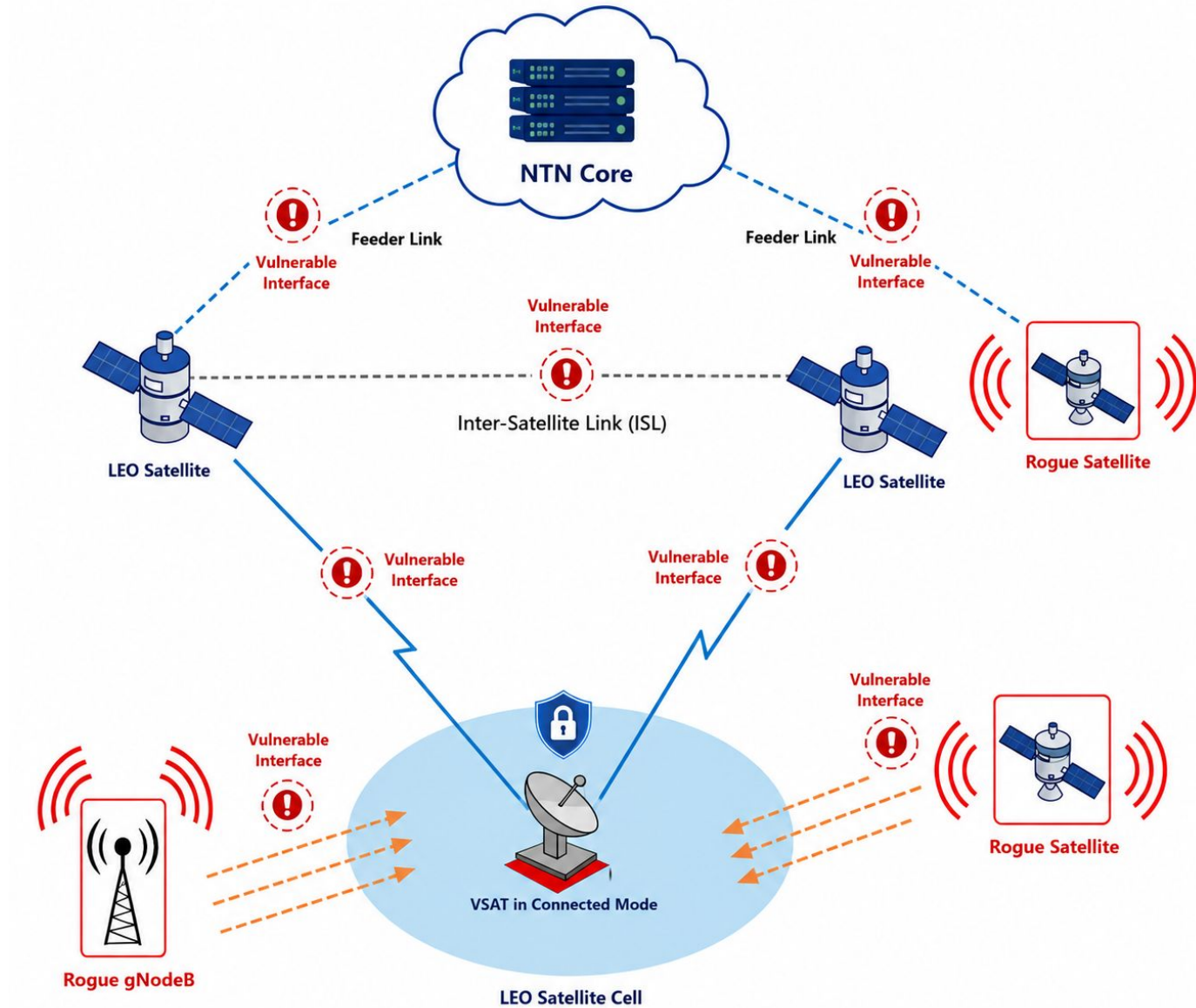
Battery-Drain Attacks

Resource exhaustion attempts designed to permanently disable critical space infrastructure.

Limited Servicing

Use of legacy components combined with slow physical maintenance and delayed remote patching cycles.

NTN attack surfaces



Threat areas:

THREAT AREA 01



**Air Interface Vulnerabilities and
Signal-Based Attacks**

THREAT AREA 02



**Rogue Satellites and
Space-Based Threats**

THREAT AREA 03



**Core Network and Cloud
Security**

THREAT AREA 04



**AI- and Quantum-Enabled
Threats**

THREAT AREA 05



**Data Security, Privacy, Trust,
and Confidence**

Are there any additional threat areas and attack surfaces?

Questions

- 01 What are the unique challenges posed by the NTN networks?
- 02 How are these challenges different compared to the terrestrial wireline/wireless networks?
- 03 What is the threat landscape?
- 04 What are the unique security mechanisms that need to be developed for the NTN networks?
- 05 What are the most urgent research gaps?

Are there any additional questions for the discussion?